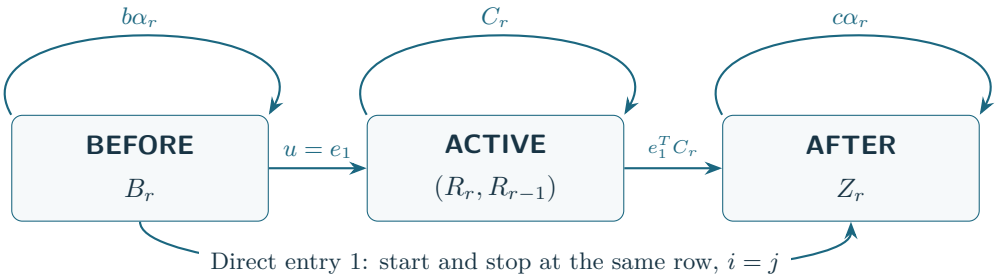
## Keep the exact source and both ends of the switch

Four states: one before, two active, one after




$$(-1)^d H_{V,\mathrm{LEFT}} \quad = \quad A^{\alpha}_{\mathrm{rest}}\, c^{d-M} Z_M$$

Prefix, suffix, omitted-half zeros and raw sign stay exact.

Full numerator: $(-1)^d H_V \quad = \quad Z_d \ (M \quad = \quad d,\ A^{\alpha}_{\mathrm{rest}} \quad = \quad 1).$

Below: restrict to disjoint start/stop windows; the path $i = j$ is excluded.

### An analytic gauge separates one regular switch

Generic triangular blocks $A, C$: full Sylvester separation is required.

**Boundary term 1**

$H_n(C_n \cdots C_1)$

Kept exactly

$+$

**Boundary term 2**

$-(A_n \cdots A_1)H_0$

Kept exactly

$+$

**Residual switches**

$\sum A \cdots R_h \cdots C$

$\|R_h\| \ \le \ Ce^{-\kappa N}$

The residual is multiplied by the original switch products.
It is not a claim that one diagonal branch determines the whole rate.

### Proved on strict regular windows

Both boundary actions lie below the interior window maximum.

Outside the local phase alignment: $|W_N| \quad \le \quad Ce^{-\varepsilon N} E_N.$

$E_N$ is the original positive window envelope.

### One shared face torus

Inconsistent $0/\pi$ incidence constraints force a mismatched port.

The original Cauchy coefficient gains $e^{-\varepsilon N}$
relative to the same product envelope and unchanged face targets.

### Global volume comparison remains open for this route

No bound of this envelope by $e^{N\Phi_D}$ is proved here.

Compatible phase systems and irregular or protected boundary windows
still require their own source estimates.